

A Strengthened Tree Banach–Saks Partial Answer to Braga’s WSC Coarse Embedding Problem

FA/Banach discovery run `fa_banach_001`

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Status

This packet concerns Bruno de Mendonca Braga’s paper *On asymptotically uniformly smoothness and nonlinear geometry of Banach spaces*, arXiv:1808.03254 [3]. It incorporates the supplied report

`evidence/supplied_braga_weak_p_banach_saks_note_2026_06_22/`.

The result is a strengthened partial answer, likely valid pending human review. It is not a full solution of Problem 8.6.

Source Problem

Problem 8.6 of the source paper asks:

Let X and Y be Banach spaces. Assume that Y has the weak p -Banach–Saks property, for some $p \in (1, \infty)$. If X coarse Lipschitzly embeds into Y by a weakly sequentially continuous map, does it follow that X has the weak p -Banach–Saks property?

coarse Lipschitz embedding from X into Y ?

At last, we ask whether a version of Theorem 6.10 holds for the weak p -Banach–Saks property. Precisely, we ask the following.

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Classification

The packet proves positive results under stronger or additional hypotheses. The supplied report also explains why the original weak- p hypothesis does not feed directly into the nonlinear telescoping proof: the construction creates weakly null trees, while weak p -Banach–Saks controls only one weakly null sequence at a time. Braga’s own examples show that this tree-versus-sequence gap is genuine [3, Theorem 7.1].

The previous packet’s tree- p permanence theorem is correct. The report adds three local repairs: the signed stabilization citation is Braga’s Proposition 6.3 rather than Proposition 6.5; normalization of semi-normalized weakly null increments preserves weak nullity; and the rescaled translated map used in the proof remains weakly sequentially continuous.

Definitions

For a map $f : X \rightarrow Y$ between Banach spaces, set

$$\omega_f(t) = \sup\{\|f(x) - f(y)\| : \|x - y\| \leq t\}, \quad \rho_f(t) = \inf\{\|f(x) - f(y)\| : \|x - y\| \geq t\}.$$

The map is a coarse embedding if $\omega_f(t) < \infty$ for all t and $\rho_f(t) \rightarrow \infty$. It is coarse Lipschitz if there are $a, b > 0$ and $A \geq 0$ such that

$$a\|x - y\| - A \leq \|f(x) - f(y)\| \leq b\|x - y\| + A \quad (x, y \in X). \quad (1)$$

It is weakly sequentially continuous if $x_n \rightharpoonup x$ implies $f(x_n) \rightharpoonup f(x)$.

A Banach space Z has weak p -Banach–Saks constant C if every weakly null sequence $(z_n) \subset B_Z$ has an infinite $M \subset \mathbb{N}$ such that

$$\left\| \sum_{j=1}^k z_{n_j} \right\| \leq Ck^{1/p} \quad (n_1 < \dots < n_k, n_j \in M).$$

It has tree- p -Banach–Saks constant C if every normalized weakly null tree of height k has a branch satisfying the same $Ck^{1/p}$ upper bound for the branch sum. Tree- p implies weak p by applying the tree property to the tree generated by a sequence [3, Remark 6.4].

Proposition 1 (Signed stabilization). *If Z has tree- p -Banach–Saks constant C , then every normalized weakly null tree of height k has a branch on which all signed branch sums have norm at most $Ck^{1/p}$.*

This is Braga’s Proposition 6.3 [3]. It follows by applying the tree property to finitely many levelwise sign changes and then using Ramsey stabilization.

Lemma 1 (Sign-cube convexity). *If $e_1, \dots, e_k$ are vectors in a normed space and $t_1, \dots, t_k \in [0, 1]$, then*

$$\sum_{j=1}^k t_j e_j \in \text{conv}\left\{ \sum_{j=1}^k \varepsilon_j e_j : \varepsilon_j \in \{-1, 1\} \right\}.$$

Consequently, if all signed sums on the right have norm at most K , then $\|\sum_j t_j e_j\| \leq K$.

Proof. Choose independent signs ε_j with expectation t_j . Then $\sum_j t_j e_j$ is the expectation of $\sum_j \varepsilon_j e_j$, hence lies in the convex hull. $\square$

Proof Intuition

The nonlinear map is applied to partial sums along a weakly null tree. The increments

$$f(r(u_s + x_{s \smallfrown n})) - f(ru_s)$$

are weakly null successor sequences because f is weakly sequentially continuous. Compression gives a lower bound on their norms and expansion gives an upper bound. After normalization, tree- p -Banach–Saks gives a branch where all signed sums are small. The sign-cube lemma then handles the unequal increment lengths, and telescoping converts the increment sum into the single endpoint difference controlled from below by ρ_f .

Modulus Transfer

Theorem 1 (Modulus transfer). *Let $1 < p \leq \infty$. Suppose that Y has tree- p -Banach-Saks constant C , and let $f : X \rightarrow Y$ be a weakly sequentially continuous coarse embedding. Fix $r > 0$ such that $\rho_f(r) > 0$. Then every normalized weakly null tree in X of height k has a branch $(x_1, \dots, x_k)$ satisfying*

$$\rho_f \left(r \left\| \sum_{j=1}^k x_j \right\| \right) \leq C \omega_f(r) k^{1/p}, \quad (2)$$

with the convention $k^{1/\infty} = 1$.

Proof. After translation assume $f(0) = 0$. For a nonempty node s , let

$$u_s = \sum_{t \preceq s} x_t, \quad y_s = f(ru_s) - f(ru_{s^-}).$$

For each fixed predecessor, the successor sequence of increments is weakly null. Since adjacent partial sums are distance r apart,

$$0 < \rho_f(r) \leq \|y_s\| \leq \omega_f(r) < \infty.$$

Thus $e_s = y_s / \|y_s\|$ is a normalized weakly null tree. Apply Proposition 1 to (e_s) . On the selected branch, put

$$t_j = \frac{\|y_j\|}{\omega_f(r)} \in [0, 1].$$

By Lemma 1,

$$\left\| \sum_{j=1}^k y_j \right\| = \omega_f(r) \left\| \sum_{j=1}^k t_j e_j \right\| \leq C \omega_f(r) k^{1/p}.$$

The left side is

$$\left\| f \left(r \sum_{j=1}^k x_j \right) - f(0) \right\|.$$

The definition of ρ_f gives (2). □

Corollary 1 (Coarse Lipschitz tree permanence). *Let $1 < p < \infty$. If Y has tree- p -Banach-Saks constant C and $f : X \rightarrow Y$ is weakly sequentially continuous and satisfies (1), then X has tree- p -Banach-Saks. More precisely, for every $r > A/a$, every normalized weakly null tree has a branch with*

$$\left\| \sum_{j=1}^k x_j \right\| \leq \frac{C(br + A) + A}{ar} k^{1/p}.$$

Letting $r \rightarrow \infty$, the constants can be taken arbitrarily close to bC/a . Hence X also has weak p -Banach-Saks.

Proof. Use $\omega_f(r) \leq br + A$, and apply the lower affine estimate in (1) to the endpoint $r \sum_j x_j$. Since $k^{1/p} \geq 1$, the additive A is absorbed as displayed. The final claim follows from tree- $p \Rightarrow$ weak- p . □

Corollary 2 (Power compression). *Under the hypotheses of Theorem 1, suppose $\rho_f(t) \geq ct^\alpha - d$ for some $c > 0$, $d \geq 0$, and $\alpha > 0$. Then every normalized weakly null tree in X has a branch satisfying*

$$\left\| \sum_{j=1}^k x_j \right\| \leq \frac{(C\omega_f(r) + d)^{1/\alpha}}{rc^{1/\alpha}} k^{1/(\alpha p)}.$$

If $\alpha p > 1$, this gives tree- (αp) -Banach-Saks in X .

Remark 1 (Endpoint). *For $p = \infty$, Theorem 1 gives a branch whose sums are uniformly bounded independently of the height. Thus tree- ∞ -Banach-Saks passes under weakly sequentially continuous coarse embeddings.*

Weak p Under Additive Defect

The original weak- p hypothesis can be used directly if the embedding is sufficiently additive on finite sums. After translating f , write $\tilde{f} = f - f(0)$, and define

$$\text{Def}_f(k, r) = \sup_{x_1, \dots, x_k \in B_X} \left\| \tilde{f}\left(r \sum_{j=1}^k x_j\right) - \sum_{j=1}^k \tilde{f}(rx_j) \right\|.$$

Theorem 2 (Defect criterion). *Let $1 < p < \infty$. Suppose that Y has weak p -Banach-Saks constant C , and let $f : X \rightarrow Y$ be weakly sequentially continuous and satisfy (1). If there are $D \geq 0$ and scales $r_k \geq 1$ such that*

$$\frac{\text{Def}_f(k, r_k)}{r_k} \leq Dk^{1/p} \quad (k \in \mathbb{N}),$$

then X has weak p -Banach-Saks, with constant at most

$$\frac{C(b + A) + D + A}{a}.$$

Proof. Let $(x_n) \subset B_X$ be weakly null and fix k . Define

$$g_k(x) = \frac{\tilde{f}(r_k x)}{r_k}.$$

Then g_k is weakly sequentially continuous, $g_k(0) = 0$, and $\|g_k(x_n)\| \leq b + A$. Apply weak p -Banach-Saks in Y to $(g_k(x_n))$, obtaining an infinite M such that

$$\left\| \sum_{j=1}^k g_k(x_{n_j}) \right\| \leq C(b + A)k^{1/p} \quad (n_1 < \dots < n_k, n_j \in M).$$

The defect assumption gives

$$\left\| g_k\left(\sum_{j=1}^k x_{n_j}\right) \right\| \leq (C(b + A) + D)k^{1/p}.$$

The lower affine estimate for g_k , with $A/r_k \leq Ak^{1/p}$, gives the displayed constant. $\square$

Corollary 3 (Bounded or sublinear additive defect). *The conclusion of Theorem 2 holds if either*

$$\sup_{u,v \in X} \|f(u+v) - f(u) - f(v) + f(0)\| < \infty$$

or, with

$$\delta_f(R) = \sup_{\|u\|, \|v\| \leq R} \|f(u+v) - f(u) - f(v) + f(0)\|,$$

one has $\delta_f(R)/R \rightarrow 0$.

Proof. Bounded binary defect gives $\text{Def}_f(k, r) \leq (k-1)D_0$; take $r_k = k$. For sublinear defect, use

$$\text{Def}_f(k, r) \leq (k-1)\delta_f(kr)$$

and choose r_k so large that the ratio is at most $k^{1/p}$. $\square$

Why This Still Does Not Settle Problem 8.6

Weak sequential continuity makes successors of each fixed tree node weakly null, but it does not make an enumeration of all increments weakly null, because the base point changes from node to node. Weak p -Banach–Saks can control a single sequence such as $f(rx_n) - f(0)$; the coarse lower estimate instead sees $f(r \sum_j x_{n_j}) - f(0)$. The difference between these two quantities is exactly the additive defect isolated above.

The supplied report also checked natural counterexample routes. The reflexive Kalton–Peck space Z_p remains a possible source obstruction, but no suitable weakly sequentially continuous embedding into a weak- p target was found. Braga’s Baire-sum construction gives weak- p spaces with very bad trees, but no coarse Lipschitz weakly sequentially continuous embedding realizing the bad tree behavior was obtained. These failures are not counterexamples.

Literature and Novelty Status

The supplied report records searches through June 21, 2026 for the exact problem number, weak- p Banach–Saks phrases, weakly sequentially continuous coarse Lipschitz embeddings, and forward citations of the source. Later nearby work develops tree- p -Banach–Saks and Hamming concentration properties [5], studies asymptotic smoothness and strict hierarchies [4], and gives further concentration examples [6]. No located paper states a proof or a counterexample to Problem 8.6.

Thus the packet should be reviewed as a strengthened partial candidate, not as a full solution.

References

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- [2] B. de Mendonca Braga, *Nonlinear weakly sequentially continuous embeddings between Banach spaces*, Int. Math. Res. Not. IMRN (2020), no. 18, 5506–5533.
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- [4] R. M. Causey, A. Fovelle, and G. Lancien, *Asymptotic smoothness in Banach spaces, three-space properties and coarse Lipschitz geometry*, arXiv:2110.06710, version 2, 2022.
- [5] A. Fovelle, *Hamming graphs and concentration properties in Banach spaces*, arXiv:2106.04297, version 3, 2023.
- [6] A. Fovelle, *Asymptotic smoothness, concentration properties in Banach spaces and applications*, arXiv:2311.15900, version 2, 2024.